SMOOTHNESS OF A RING MAP IS PRESERVED BY LOCALIZING THE SOURCE AND BY INVERTING AN ELEMENT OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This is the first card this lane authors from stacks-algebra, the source newly assigned to it, and the first card in the L10000-L11999 range.

Editorial note. The staged candidate list distributed with this source records ext_refs = 0 for this label, but the proof does reference two same-chapter lemmas; both are recorded as navigation stubs whose tags were looked up by the same-chapter suffix rule, and the cotangent-complex argument stays inside the counted span.

Editorial note. Assembly history: r001 was rejected with "assembly differs from reviewed statement locators" because the hand-written body_spans carried shortened labels while the contract (prepared_delivery.py line 249) requires body_spans to equal the top-level statement/proof spans entry by entry; r002 rebuilds body_spans from those spans so the two are identical. No span coordinate, statement or proof text changed.

Allowed standard prerequisites (editorial).

- The naive cotangent complex of a ring map. Same chapter, Lemma labelled lemma-localize-NL; invoked at proof line 37339. Identifies the naive cotangent complex of S_g over R with the base change of that of S over R.
- Localisation of the naive cotangent complex in the base. Same chapter, Lemma labelled lemma-NL-localize-bottom; invoked at proof line 37345. Gives the analogous statement for the second assertion, after inverting an element of the base.

Source reference. Original lemma 00S7: sources/stacks-algebra/pinned-original-source.tex, source lines 35909–35909. Navigation pointer for \ref{lemma-localize-NL} used inside the counted proof (line 35909); the cotangent-complex argument that consumes it is retained in the span.

Source reference. Original lemma 07BS: sources/stacks-algebra/pinned-original-source.tex, source lines 35807–35807. Navigation pointer for \ref{lemma-NL-localize-bottom} used inside the counted proof (line 35807); the cotangent-complex argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a smooth ring map. Any localization S_g is smooth over R. If $f \in R$ maps to an invertible element of S, then $R_f \to S$ is smooth.*

Proof. By Lemma 00S7 the naive cotangent complex for S_g over R is the base change of the naive cotangent complex of S over R. The assumption is that the

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naive cotangent complex of S/R is $\Omega_{S/R}$ and that this is a finite projective S-module. Hence so is its base change. Thus S_g is smooth over R.

The second assertion follows in the same way from Lemma 07BS. $\square$